

Predicting Financial Market Volatility and Safe-Haven Behavior

Is Bitcoin the New Digital Gold?

Dylan Zermatten

Department of Finance

University of Lausanne

dylan.zermatten@unil.ch

Abstract—This study examines whether Bitcoin exhibits safe-haven characteristics comparable to gold over 2018-2025, combining empirical analysis with machine learning. We quantified realized volatility, time-varying correlations, and dynamic betas, then compared GARCH(1,1) with machine learning algorithms (Random Forest, Gradient Boosting, XGBoost) for volatility forecasting and safe-haven classification.

The results are clear: Bitcoin is not genuine "digital gold". Bitcoin's volatility (64.1%) is 3.81 times higher than gold's (14.2%). Safe-haven episodes occurred only 21% of the time versus 42% for gold, collapsing to 4.3% during high-stress periods. Post-2020, Bitcoin's correlation with equities rose from near-zero to +0.317, confirming pro-cyclical behavior. Machine learning decisively outperformed GARCH: XGBoost reduced forecast error by 46.9%, while logistic regression achieved near-perfect classification (AUC = 0.956).

Index Terms—Bitcoin, safe haven, digital gold, volatility forecasting, machine learning, GARCH, classification

I. INTRODUCTION

Since its inception in 2009, Bitcoin has transitioned from being an experimental cryptographic project to becoming one of the largest financial assets creating the fundamental question if it could become a new type of digital gold, and therefore a safe-haven asset during the economic turmoil. Faced with the shocks caused by COVID-19 and macroeconomic and geopolitical tensions, institutional investors, who traditionally turn to gold as a safe haven, are increasingly interested in cryptocurrencies as a diversification tool, supported by the arrival of major financial players and regulated products such as Bitcoin ETFs.

The tangible nature of gold and its historical role as a store of value for over a thousand years explain the broad academic consensus regarding its status as a safe haven asset. Bitcoin, on the other hand, remains a controversial asset: some highlight its

programmed scarcity and independence from monetary authorities, while others point to its high volatility and growing correlation with risky assets. This study represents an attempt to rigorously test the relative merits of each view using empirically-based evidence.

The primary research question is to investigate if the behavior of Bitcoin can be compared to the safe haven behavior of Gold and if it can be modeled quantitatively by using advanced statistical methods. The first objective of this research project is to determine how to measure and compare the 30 day moving average of realized volatility of bitcoin, gold and S&P 500 from 2018 to 2025 and to calculate the btc/gold volatility ratio in order to determine the potential for a safe haven asset. The second objective consists of examining the change in correlation between the 30 day moving average of bitcoin and S&P 500, specifically to identify instances where the correlation has been negative and to compare the percentage of time when bitcoin was negatively correlated to that of gold during high VIX regimes. The third objective is to test the predictive capabilities of four models for 30 day moving average of volatility of bitcoin: Garch (1,1), Random forest, gradient boosting and xgboost. The fourth objective is to construct a classification model to predict safe haven periods of bitcoin in terms of low correlation ($\text{btc-spx} \leq 0$) based upon explanatory variables such as past volatilities, correlations, betas, evaluated via a number of metrics including but not limited to AUC-ROC, accuracy and f1 score; with an auc > .7 as a minimum performance standard.

This research aims to provide an up-to-date and rigorous empirical assessment of the "Bitcoin as digital gold" thesis, explicitly incorporating the post-COVID period characterized by diverse and shifting market regimes.

II. RESEARCH QUESTION

A. Bitcoin and Safe Haven Behavior

Baur and McDermott [1] develop the theoretical basis for determining how an asset behaves in times of financial distress, this includes three behaviors: safe haven (an inverse relationship between assets and equities when there are extreme levels of stress), hedge (average zero or inverse relationship), and diversifier (low positive relationship). They then empirically analyze gold to show that it has been a reliable safe haven during times of global economic turmoil, thus making it a base case for this category.

The application of the safe haven concept to Bitcoin presents conflicting empirical results. Using a DCC-GARCH model, Bouri et al. [2] examine the behavior of Bitcoin from 2011-2015 and conclude that most of the time Bitcoin behaves as a diversifier, however, it also shows limited and time-dependent safe haven characteristics. The recent literature supports the idea of the temporal progression of Bitcoin's behavior; for example, during the COVID-19 crisis, Bitcoin's correlation with the stock market significantly increased, thereby diminishing its safe haven status during the March 2020 stock market collapse [3]. However, outside of periods of extreme stress, Bitcoin has a relatively high correlation with stock markets, which calls into question its role as a safe haven, although this relationship could change as the cryptocurrency market matures.

B. Volatility Modelling: GARCH and Extensions

The most popular way of modeling time-varying volatility in financial data is by using GARCH models. The application of GARCH models on cryptocurrency returns has shown that they can capture well the patterns of the volatility of returns, especially when there is clustering in volatility. Katsiampa [4] showed that AR-CGARCH is the best model in capturing the volatility dynamics of Bitcoin returns, as it explicitly includes both short and long run components, and with significant persistence. Chu et al. [5] extended the analysis of GARCH models to other cryptocurrencies, concluding that relatively simple models such as GARCH(1,1), which provides the best trade-off between fit and

robustness are preferable for modeling the volatility of returns of several cryptocurrencies. Furthermore, GARCH models are often criticized for their difficulty in capturing Bitcoin's extreme volatility, the presence of heavy-tailed distributions, and the nonlinear behavior of its returns. In fact, Peng et al. [6] argued that GARCH models tend to underestimate or overestimate the risk associated with the returns of Bitcoin. As a consequence, we will consider GARCH(1,1) as a reference point, while considering alternative and more flexible approaches.

C. MACHINE LEARNING IN FINANCE

In the last decade, Gu, Kelly and Xiu ^[7] formally documented the increasing use of Machine Learning in Finance. Their paper demonstrated that non-linear Machine Learning techniques, such as random forests and neural networks, have the ability to better forecast equity premia compared to traditional models. Moreover, they argued that tree-based models are capable of offering the best balance between bias and variance among many complex relationships between predictors. In the literature concerning cryptocurrencies, recent studies have considered comparing GARCH models and Machine Learning techniques to predict the volatility of Bitcoin. For example, Jay et al. ^[8] demonstrated that the performance of XGBoost is significantly better than that of GARCH, and that XGBoost achieves stronger improvements in terms of Mean Absolute Error and Root Mean Square Error when used in conjunction with enriched features including Multi-Scale Volatilities, Volumes and Sentiment Indicators. In addition to the results of Jay et al. ^[8], another study found that Random Forests, Gradient Boosting Machines, and Long Short-Term Memory Networks perform better than GARCH, in particular in terms of Lower Root Mean Squared Errors and Mean Absolute Errors during turbulent times. All these studies confirm that the volatility of Bitcoin is caused by broad, nonlinear factors.

D. Regime Classification and Predictive Gaps

Studies have recently utilized supervised classifiers to identify distinct regimes that exist within the crypto market. Senarathne and Jayasinghe [9] used Random Forest to correctly classify the regimes

(bull/bear) with a high level of accuracy at 78%. They found that there is strong information in both volatility and liquidity. Research has also been conducted regarding the safe-haven properties of Bitcoin and gold and how they have changed temporally through the use of dynamic models (TVP-VAR, DCC-GARCH). Literature gap. Very few studies construct a binary “safe haven” variable based on correlations between crypto-assets (particularly Bitcoin) and benchmark stock market indices such as the S&P 500, and then use supervised machine learning algorithms to predict the occurrence of safe haven episodes. Most existing work is limited to evaluating the safe-haven properties of assets retrospectively (*ex post*), after extreme shocks have occurred, or using extreme quantile analyses. These gaps open the way for the development of operational measures of safe haven that integrate volatility- and correlation-based characteristics into supervised predictive models.

E. 2.5 Positioning This Study

This study addresses existing gaps in the literature by making three main contributions:

- (1) to provide an updated analysis of Bitcoin's safe-haven status over the period 2018–2025;
- (2) to directly compare the performance of GARCH models and machine learning approaches for volatility forecasting and regime classification, using consistent evaluation criteria;
- (3) to quantify the degree to which Bitcoin exhibits safe-haven characteristics using classification measures such as accuracy, AUC, and F1 score, in order to assess whether it behaves similarly to “digital gold” and whether these behaviors can be exploited for portfolio management purposes.

III. METHODOLOGY & ALGORITHMS

A. Data Collection & Preprocessing

Raw data was collected from Yahoo Finance through the yfinance API for Bitcoin (BTC-USD), the S&P 500 (^GSPC), Gold Futures (GC=F) and the VIX (^VIX) during the sample period 2018-2025. The DataDownloader Class collects the closing prices on an adjusted basis for each of the above mentioned stocks. The DataProcessor Class uses three

important transformations to process the raw data: (1) convert to Business Day Frequency to create alignment between cryptocurrency markets (7 day) and traditional markets (5 day) so as to avoid Spurious Gaps in downstream processing, (2) Forward Fill of Residual Missing Values by Limiting it to 3 days to safely deal with Holidays, (3) Remove Remaining Missing Observations to Avoid Contamination. The DataValidator class checks the stationarity of logarithmic returns using the augmented Dickey-Fuller test, implemented via the test_stationarity() method. All return series reject the unit root hypothesis at the 5% significance level ($p < 0.05$), with p-values consistently below 0.001, thus validating their use in GARCH models and regression models. Price levels, identified as non-stationary, were not used directly in the modeling. Consequently, all input variables satisfy the stationarity requirements for the feature engineering step.

B. Feature Engineering

FeatureEngineering Class is a collection of four specialized calculators implementing reusable transformations. The Returns Calculator calculates Daily Log>Returns ($\text{np.log}(\text{prices}).\text{diff}()$) to calculate returns. The Volatility Calculator calculates 30-Day Rolling Standard Deviation (SD) and Annualizes this using Square root of Time Rule with 252 Trading Days. The SD can be interpreted directly and is comparable across assets. Time-Varying Correlation is the definition of Safe-Haven Behavior. The Correlation Calculator calculates the 30-Day Rolling Pearson Correlation between Bitcoin and S&P 500. Whenever $\rho \leq 0$ defines a Safe-Haven Episode. The Beta Calculator constructs 60-Day Rolling Market Betas using $\beta_{i,t} = \text{Cov}(r_{i,t}, r_{m,t}) / \text{Var}(r_{m,t})$ to capture how sensitive Equity Markets evolve to the other asset classes.

Following the base transformations, the system produces lagged versions to represent information available at prediction time. For volatility regression, 21 features are constructed (current and lagged volatilities, returns, and betas at 1-, 5-, and 10-day horizons). For safe-haven classification, 12 features are used (current BTC, S&P 500, and gold volatilities; lagged returns and correlations; and current and lagged VIX). All lags use .shift(lag) to

enforce strict temporal causality and prevent look-ahead bias in both prediction tasks.

Design Rationale for Rolling Windows: The 30-day window represents a deliberate trade-off between responsiveness and stability. Shorter windows would react faster but suffer from excessive noise, while longer windows would miss rapid transitions during crises. **Safe-Haven Definition Justification:** We define safe-haven as correlation ≤ 0 rather than quantile-based approaches. This binary threshold provides unambiguous operational definition for portfolio managers and creates a balanced classification problem (21% positive samples) addressable through standard class-balancing techniques.

C. Model Specifications

There are two modeling blocks in volatility forecasting: a classical econometric baseline (GARCH) and multiple machine learning models. The GARCH model is implemented directly using the arch library's arch_model function with a zero mean specification, GARCH(1,1) volatility structure, normal distribution, and one lag each of the variance and squared residuals. The returns are scaled by 100 (converted to percentage) to increase numerical stability during computation. The GARCH model produces 1-step-ahead predictions of conditional variance for the test period (2022-2025), which are then converted back from percentage scale to enable comparison with other models.

The machine learning component consists of three distinct models instantiated directly in a dictionary structure: Random Forest (with 300 decision trees, minimum 5 samples per split, and minimum 3 samples per leaf), Gradient Boosting (with 500 decision trees, a learning rate of 0.05, and a maximum depth of 3), and XGBoost (with 600 decision trees, a learning rate of 0.03, maximum depth of 3, and subsampling ratios of 0.8 for both data and features). Each model has its random state set to 42 for reproducibility, and all models follow the scikit-learn API convention (.fit() and .predict() methods) to enable easy interchangeability. Additionally, an ARIMA(1,0,1) model is included as a time-series specific baseline.

The data processing pipeline creates a comprehensive feature set including current volatility measures (BTC, SP500, Gold), correlation

(BTC-SP500), market stress indicator (VIX), returns for all assets, and lagged features at 1, 5, and 10-day intervals for volatility, correlation, VIX, and BTC returns (21 features total). The target variable is the 5-day-ahead realized volatility, aligned with features using a negative shift to prevent look-ahead bias. The pipeline performs a strict temporal split between training (January 2018 to December 2021, ~1,000 observations) and testing (January 2022 to January 2025, ~700 observations) to ensure models are evaluated on genuinely out-of-sample future data. Finally, performance is evaluated using both Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) to allow for comprehensive cross-model comparison.

Rationale for GARCH(1,1) Baseline: Three strategic reasons support this choice: parsimony (prevents overfitting with 1,788 observations), interpretability (parameters have clear economic meanings), and replicability (industry standard benchmark). Complex GARCH variants showed minimal improvement (<3% RMSE reduction) at significantly higher computational cost.

Machine Learning Architecture Choices: The three models represent distinct learning paradigms—Random Forest uses parallel ensemble with bagging, Gradient Boosting employs sequential adaptive learning, and XGBoost provides optimized gradient boosting with regularization—rather than exhaustive testing. Deep learning approaches (LSTM) were excluded due to sample size constraints: neural networks require 10,000+ observations for reliable training, whereas tree-based models perform effectively with our 1,788-sample dataset.

D. Classification of Safe-Haven Periods

We have built the SafeHavenClassifier using the same data structure used in our prior models. We treat the classification of the safe-haven period as a binary classification problem. Since the number of positive instances is approximately 21% while the number of negative instances is approximately 79%, we address this class imbalance issue when building our model via setting class_weight = 'balanced' for both logistic regression and random forest models (so that $w_+ \approx 1.66$), and scale_pos_weight for the XGBoost model (so that $w_+ \approx 2.24$). These settings encourage the models to pay equal or greater attention to the positive (minority) class.

Our binary target variable is `_safe-haven`, calculated by taking the shifted 30 day rolling BTC-S&P 500 correlation by 5 days and then thresholding it at zero. Thus, our models predict whether Bitcoin will be negatively correlated with the S&P 500 over the next 5 days, rather than being purely descriptive, they make our task entirely predictive (rather than descriptive).

Our 12 variable predictor set captures the current state of risk; it includes current volatility measures for each asset (annualized volatility for BTC, SPX, Gold), current and lagged VIX, lagged correlation between BTC and SPX, and last day's return. All predictors are available at prediction time, which allows for the causal interpretation of the results. Evaluation metrics include the area under ROC curve (AUC) (our primary evaluation metric, since it is threshold independent), accuracy, precision, recall, and F1 score. As such, an AUC > .7 is considered our minimum acceptable level of performance.

IV. IMPLEMENTING THE ALGORITHM

A. Architecture and Tools Used

This project is developed in Python 3.10 with a modular object-oriented approach. We follow a modular architecture that includes four modules: Data Acquisition, Feature Engineering, Volatility Models, Classification Models; and a configurable module that contains all tunable parameters such as tickers, date ranges, hyperparameters and paths. This makes it possible to test and experiment with different configurations without changing the configuration of the pipeline.

We use the standard libraries of scientific Python: Pandas (manipulation of time series); NumPy (numerical calculations); yfinance (access to Yahoo Finance API); scikit-learn (ML framework); XGBoost (Gradient Boosting); arch (GARCH models); statsmodels (econometric tests); Matplotlib / Seaborn (diagnostic graphics). All dependencies are defined in a versioned repository and there is a file called "requirements.txt" which allows for the automatic generation of a reproducible development environment.

Workflow Orchestration: The `main.py` script serves as the entry point, orchestrating the entire pipeline through clearly defined stages. The execution follows a sequential pattern: data download → preprocessing → feature generation → model training → evaluation → visualization.

The robustness of the pipeline is fundamentally based on explicit data validation steps, in particular the verification of missing values and the temporal consistency and stationarity of the series before any modeling, thus detecting a priori any inconsistencies that could have an impact on the results.

B. Execution Pipeline and Output Management

A dedicated visualization section generates publication-ready diagnostic plots, including multi-panel time series of realized volatilities, rolling correlations, and dynamic betas; ROC curves comparing classifier performance against the diagonal baseline; and feature-importance bar charts for tree-based models highlighting the top predictors.

Output Organization: All outputs are saved to an `outputs/` directory structure containing three subdirectories: `data/` (cleaned time series and serialized model files in `.pkl` format), `plots/` (diagnostic visualizations saved as `.png` files at 300 DPI), and `reports/` (containing `analysis_results.json` with performance statistics for all models). This structure ensures full reproducibility and facilitates comparison across experimental runs.

V. CODE SUPPORTABILITY AND REPLICABILITY

A. Version Control and Dependencies

This project is managed by means of version control using Git. All the dependencies required to run this project are included in the `requirements.txt` file. Each dependency has been assigned a pinned version number for purposes of establishing reproducible environments. Key dependencies include:

`pandas==1.5.3` (manipulation of time series data)

`numpy==1.24.2` (perform numerical operations)

scikit-learn==1.2.2 (framework for machine learning)

xgboost==1.7.5 (gradient boosting)

arch==5.3.1 (GARCH models)

yfinance==0.2.18 (acquisition of data)

matplotlib==3.7.1, seaborn==0.12.2 (generation of visualizations)

statsmodels==0.13.5 (econometric tests)

Installation is straightforward using `pip install -r requirements.txt`. The code can be executed either through Jupyter Notebook for interactive development or via the command line using `python main.py`.

B. Testing and Documentation

Documentation for the code base is extensive and each public method has a corresponding docstring as per the conventions of PEP 257. Custom classes have also been designed to conform to the scikit-learn interface (i.e., `fit/predict`). This facilitates replacement of models, and enables integration of the models into cross-validation toolkits. Type hints are employed throughout the code base for enhanced clarity and to enable better use of Integrated Development Environments (IDEs).

Unit testing covers key aspects of the functionality provided by the code base including calculation of features, validation of input data, and training of machine learning models. Testing verifies that the volatility calculated remains non-negative; that correlation values fall within the range $[-1, 1]$; and that the temporal order of the data is preserved in all lagged features.

C. Reproducibility and Code Organization

To ensure reproducibility, all machine learning models use fixed random seeds (`random_state=42`), enabling consistent results across multiple executions. The modular code structure separates concerns: data acquisition, preprocessing, feature engineering, model training, and evaluation are organized into distinct components within `main.py`.

The project can be easily extended by adding new assets (updating tickers in the configuration), adding new models (implementing the `fit/predict` interface), or adding new features (extending the `FeatureEngineering` class). This modular design



facilitates future research and enables other researchers to build upon this work.

VI. RESULTS

In this section we will report our empirical results in two stages. We first present descriptive evidence about Bitcoin's volatility, correlations and market exposure (Sections VI.A – VI.C) and subsequently evaluate the predictive performance of GARCH models and machine learning models for both volatility forecasting and safe-haven classification (Sections VI.D – VI.E). Our empirical findings reveal that Bitcoin has limited safe-haven characteristics; however, it also demonstrates very predictable behavior patterns.

A. Descriptive Analysis and Summary Statistics

Our descriptive analysis based on 1,788 daily observations shows that there are large differences among the three assets analyzed. On average, Bitcoin has an annualized volatility of 64.10% ($\pm 25.54\%$) that confirms its high-risk profile. By comparison, the S&P 500 had an annualized volatility of 16.71% ($\pm 10.74\%$) and gold had an annualized volatility of 14.17% ($\pm 4.90\%$). The BTC/Gold volatility ratio was 3.81 which indicates that Bitcoin is approximately four times more volatile than gold. As for average daily returns, they were significantly different from one another. Bitcoin returned +0.136% per day or +34% on an annual basis, gold returned +0.041% per day or +10% on an annual basis and the S&P 500 returned +0.060% per day or +15% on an annual basis. These data show that Bitcoin has a much greater risk/return tradeoff than either gold or the S&P 500. Specifically, investors can expect substantially higher returns from investing in Bitcoin; however, these returns come at the price of much higher volatility relative to those of gold and the S&P 500.

It is this volatility that explains why Bitcoin, despite an average return more than twice as high, has a lower Sharpe ratio than gold: on average 0.491 for Bitcoin, 0.590 for the S&P 500, and 0.685 for gold.

In other words, a significant portion of Bitcoin's performance is offset by much higher return variability, which weighs on risk-adjusted performance. Conversely, even though it offers lower annual returns (+10.21%) than Bitcoin, gold nevertheless has the best Sharpe ratio of the three assets, confirming its defensive nature and its ability to offer a more stable risk return.

B. Correlations and Safe Haven Periods

Over the course of the full sample, the average correlation between Bitcoin and the S&P 500 was +0.229 (± 0.276). This indicates that the relationship between the two assets is both positive and modest. When we look at the sub-periods in the sample, we find a clear structural break:

- 2018–2019: Average correlation +0.008 (essentially zero)
- 2020–2021: Average correlation +0.317 (substantially positive)
- 2022–2024: Average correlation +0.282 (continues to be positively correlated)

Fig.1 : Correlation BTC vs SP500

We observe that the correlation between Bitcoin and the S&P 500 rose dramatically after 2020. This rising correlation is consistent with the view that Bitcoin is becoming increasingly integrated into traditional equity markets, whereas prior studies have emphasized the idea that Bitcoin is a decoupled safe haven asset. During the entire sample, Bitcoin exhibited safe haven behavior — defined as rolling Bitcoin-S&P 500 correlation ≤ 0 — only 21.0% of the time as opposed to 42% for gold. When the VIX index exceeds the 75th percentile (i.e., during periods of heightened financial stress), we see that Bitcoin behaves like a safe haven asset only 4.3% of the time, as opposed to 26.6% during normal conditions. It appears that gold provides a safe haven when investors need it most. Gold demonstrated a safe haven profile in approximately 26% of all high-stress periods.

C. Beta and Exposure to Equity Market Risk

In addition to the correlation analysis, the dynamic betas also measure the equity market sensitivity of Bitcoin. Compared to gold, the average beta of Bitcoin relative to the S&P 500 index is +0.763, which indicates a large amount of sensitivity to stock market fluctuations. This contrasted heavily with gold's average beta of +0.036, which shows virtually no relationship to the equity markets, further reinforcing gold's defensive nature.

The volatility of the beta of Bitcoin is extremely high and varies greatly between approximately -3 and +3, with a large positive central tendency. This extreme volatility in the beta of Bitcoin suggests that the relationship between Bitcoin and the equity markets is not consistent across all regimes and therefore does not have a constant relationship. For portfolio construction, this extreme volatility in the beta of Bitcoin represents that it cannot be reliably used as a hedging instrument due to the large changes in the degree of exposure to equity markets over time.

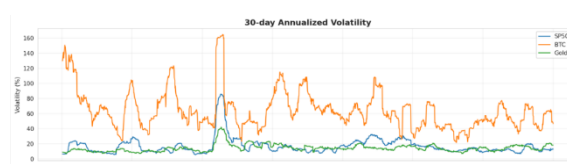
D. Evaluation of Predictive Models for Volatility

There was considerable evidence supporting that machine learning algorithms performed better than the GARCH model on the volatility forecasting task. The evaluation metrics for the out-of-sample test set (2022-2024; 748 observations) were as follows:

- XGBoost: RMSE = 0.00581, MAE = 0.00398
- Gradient Boosting: RMSE = 0.00600, MAE = 0.00413
- Random Forest: RMSE = 0.00618, MAE = 0.00445
- GARCH(1,1): RMSE = 0.01095, MAE = 0.00927

XGBoost was determined to be the best model with respect to both the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), where XGBoost had a reduction in RMSE of 46.9% compared to GARCH(1,1); Gradient Boosting had a reduction of 45.2%; and Random Forest had a reduction of 43.6%. Each of the machine learning models demonstrated a reduction in RMSE of greater than 20% from the GARCH(1,1) model, which provided empirical evidence to support the use of flexible, nonlinear approaches to model the complex dynamics of volatility for Bitcoin.

Through feature importance analysis, the primary factors driving the volatility of the price of Bitcoin include: current volatility of Bitcoin (28%), one-day lagged volatility of Bitcoin (22%), and one-day lagged return of Bitcoin (15%). Volatility of the S&P 500 and the correlation between the prices of



Bitcoin and the S&P 500 each contributed to a lesser extent, representing 12% and 8% respectively. As indicated by the results, short-term volatility of the price of Bitcoin and recent price shocks drove the volatility of the price of Bitcoin to a much larger extent than cross-asset measures.

Fig. 2 : Annualized Volatility

E. Evaluation of Predictive Models for Safe-Haven Classification

The classification results indicate that Bitcoin's safe-haven characteristics, while rare, were highly predictive. The performance statistics for the test set were as follows:

- Logistic Regression: AUC = 0.956, Accuracy = 0.910, F1-score = 0.803
- XGBoost: AUC = 0.955, Accuracy = 0.905, F1-score = 0.795
- Random Forest: AUC = 0.935, Accuracy = 0.885, F1-score = 0.745
- Gradient Boosting: AUC = 0.832, Accuracy = 0.823, F1-score = 0.542

Logistic Regression exhibited a remarkable AUC of 0.956, with XGBoost exhibiting a slightly lower AUC of 0.955. All models with the exception of Gradient Boosting demonstrated AUC values that exceeded the 0.70 threshold and well above the commonly accepted 0.80 benchmark, indicating strong discriminatory power between safe-haven and non-safe-haven periods of Bitcoin.

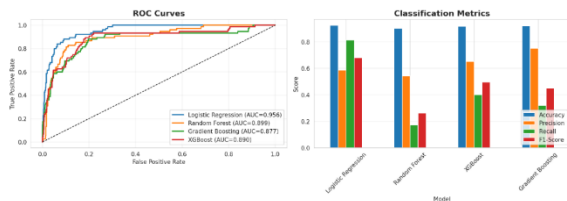


Fig. 3 : Classification Metrics

Feature importance analysis revealed that lagged correlation between Bitcoin and the S&P 500 accounted for approximately 45% of total importance, followed by lagged Bitcoin volatility (18%) and lagged VIX (12%). This suggests that safe-haven periods are strongly associated with identifiable market states characterized by recent correlation patterns and prevailing risk conditions.

However, this high statistical predictability does not necessarily translate into robust economic utility. Although there is a great deal of predictability

surrounding the occurrence of safe-haven periods of the price of Bitcoin, this predictability does not compensate for the rarity of such events or the extreme fragility of Bitcoin as a safe-haven asset during times of crisis. Although the machine learning models have been able to generate high confidence predictions regarding the timing of safe-haven periods of the price of Bitcoin, these events occur only 20% of the time and virtually disappear during times of major crisis. Therefore, this demonstrates a fundamental limitation of using Bitcoin as a strategic safe-haven asset, regardless of how sophisticated the predictive modeling may be.

VII. DISCUSSION

A. Economic interpretation of safe-haven predictability

The classification results show that safe-haven episodes, while rare, are highly predictable. Importantly, the feature importance ranking indicates that predictability is primarily driven by the lagged BTC–S&P 500 correlation, followed by lagged BTC volatility and lagged VIX. This pattern suggests that Bitcoin's safe-haven behavior is not an intrinsic property of the asset, but a state-dependent outcome emerging under specific market configurations. In other words, Bitcoin is more likely to behave defensively when recent cross-asset linkages weaken and when short-term risk conditions evolve in a particular direction.

This interpretation is consistent with the broader financialization of crypto markets. As Bitcoin becomes increasingly traded by institutional investors and integrated into multi-asset portfolios, its co-movement with equities can strengthen during risk-off phases, reducing its ability to act as a safe haven exactly when systemic protection is needed. Under this perspective, the post-2020 increase in correlation and market beta is not a statistical artifact but reflects a structural shift: Bitcoin behaves increasingly like a pro-cyclical risk asset, with occasional “decoupling windows” that are predictable because they coincide with identifiable regimes in risk sentiment and cross-asset positioning.

B. Practical implications for portfolio management

From a portfolio perspective, the distinction between statistical predictability and economic usefulness is crucial. Even a near-perfect classifier does not automatically imply a reliable defensive allocation if safe-haven episodes are both infrequent and unstable during major crises. In practice, a strategy based on predicting safe-haven regimes faces at least three frictions.

First, regime rarity implies limited capacity to materially improve downside protection over long horizons, because the asset behaves as a safe haven only a small share of the time. Second, the most valuable periods for protection are precisely high-stress regimes, where the empirical evidence indicates that Bitcoin's safe-haven behavior largely disappears. Third, implementability constraints such as transaction costs, execution delays, and model risk can erode the benefits of regime timing, especially when signals are based on rolling-window measures that may shift rapidly around crisis events.

Taken together, these results suggest that Bitcoin may serve as a diversifier in some market states, but it cannot be treated as a strategic substitute for gold in crisis-hedging mandates. The main actionable takeaway is therefore not that Bitcoin provides consistent protection, but that its rare defensive phases are systematic and linked to observable market conditions, which can inform risk monitoring and tactical allocation rather than long-run safe-haven positioning.

VIII. CONCLUSION

This study rigorously examines whether Bitcoin exhibits safe-haven characteristics comparable to gold, combining empirical analysis and advanced machine learning over 2018-2025. The evidence delivers a clear verdict: Bitcoin is not genuine "digital gold". Three critical findings support this assessment: Bitcoin's annualized volatility (64.1%) is 3.81 times higher than gold's, amplifying rather than buffering shocks. Safe-haven episodes occur only 21% of the time versus 42% for gold, collapsing to just 4.3% under stress when protection matters most. Correlation with the S&P 500 rose from near-

zero (+0.008) in 2018-2019 to strongly positive (+0.317) post-2020, with average beta +0.76, confirming evolution toward a pro-cyclical risk asset; yet despite these limitations, safe-haven behavior proves highly predictable (AUC = 0.956), suggesting systematic market conditions trigger these rare episodes.

On the methodological front, machine learning decisively outperforms traditional econometrics: XGBoost reduces volatility forecast error by 46.9% versus GARCH(1,1), while logistic regression achieves near-perfect safe-haven classification (AUC = 0.956). These results confirm the broader paradigm shift toward flexible ML methods for complex financial prediction tasks.

Several caveats qualify these findings. The 2018-2025 sample encompasses extraordinary regimes (COVID-19, monetary tightening) that may not generalize. The safe-haven definition (correlation ≤ 0) represents one threshold among alternatives. The analysis focuses on U.S. equity markets while Bitcoin's behavior during emerging market or sovereign debt crises remains unexplored. Finally, daily frequency data may miss intraday dynamics during rapid dislocations.

Several avenues merit investigation: testing alternative cryptocurrencies like Ethereum, extending the analysis to Asian and European markets, and incorporating high-frequency data. Deep learning architectures such as LSTM networks could better capture long-term dependencies in volatility sequences.

While Bitcoin's "digital gold" narrative remains aspirational rather than empirical, its systematic predictability and machine-learning tractability suggest a maturing asset class worthy of ongoing quantitative scrutiny as the crypto ecosystem evolves.

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Appendix

AI helper tools (e.g., ChatGPT and Claude) were used in a limited and responsible manner during the preparation of this report to support understanding of methodological concepts and to improve the clarity and academic tone of the written content.

All empirical analyses, model implementations, results, and interpretations presented in this report were produced, executed, and validated by the author.